

Binary Numbers and Bit Manipulation

 33_{10} $= 33_2 =$

$33/2 = 16 \% 1$

$16/2 = 8 \% 0$

$8/2 = 4 \% 0$

$4/2 = 2 \% 0$

$2/2 = 1 \% 0$

$1/2 = 0 \% 1$

100001

-33

↳ 1 0 1 0 0 0 0 1

signed bit

0 → +ve
1 → -ve

two's complement

33 → 00100001

11011110

$\text{binary}(5) + \text{binary}(7) = \text{binary}(12)$

$$\begin{array}{r} 00100001 \\ 11011110 \\ \hline 11011111 \end{array} \xrightarrow{+1}$$

→ two's complement of -33

$\text{binary}(5) - \text{binary}(7)$

$= \text{binary } 5 + \text{two's complement of } (7) = \text{num}$

$= \text{two's complement of num}$

$= \text{binary}(2)$

and &

xor ⊕

left shift <<

or |

not ~

right shift >>

↳ logical right shift

↳ Arithmetic right shift